

Unit 4: Trigonometric Functions

Precalculus

Student

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4.1 Angles, Radian, and Degree Measure

Objectives

- Describe angles in standard position.
- Distinguish positive (counterclockwise) from negative (clockwise) rotation.
- Define and find coterminal angles.
- Convert angles between degrees and radians.
- Find complementary and supplementary angles in degrees and radians.
- Locate radian angles not in terms of π on the unit circle.

Standard Position

An angle is in _____ when its vertex is at the _____ of a coordinate system, and its _____ lies along the positive x -axis.

The ray that has been rotated from the initial side to form the angle is the _____.

Direction of Rotation

If the terminal side is rotated _____, the angle is _____.

If the terminal side is rotated _____, the angle is _____.

Coterminal Angles

Two angles are _____ if they share the same terminal side. To find a coterminal angle, we add or subtract full revolutions (_____ in degrees, or _____ in radians).

$$\theta_{\text{coterminal}} = \theta \pm 360^\circ k \quad \text{or} \quad \theta_{\text{coterminal}} = \theta \pm 2\pi k \quad \text{for } k \in \mathbb{Z}$$

Example 1

Find two angles—one positive and one negative—that are coterminal with

$$\theta = \frac{\pi}{12}.$$

Degree-Radian Conversion

To convert degrees to radians, multiply by _____.

To convert radians to degrees, multiply by _____.

Example 2–4

Convert each angle measure.

2) Convert 130° to radian measure. Leave your answer in exact form.

3) Convert $-\frac{3\pi}{10}$ radians to degree measure.

4) Convert 2 radians to degree measure. Round to the nearest hundredth of a degree.

Complementary and Supplementary Angles

Two positive angles are _____ if their sum is _____ ($\pi/2$).

Two positive angles are _____ if their sum is _____ (π).

If an angle is already larger than the sum target, the complement or supplement _____.

Example 5–6

If possible, find the complement and supplement of each angle.

5) 148°

6) $\frac{2\pi}{5}$

Radians Without Pi

When an angle is given as a plain number (e.g., $\theta = 3$), it is in _____. To determine its location on the coordinate plane, compare it to the decimal approximations of the quadrant boundaries:

$$\frac{\pi}{2} \approx 1.57 \quad \pi \approx 3.14 \quad \frac{3\pi}{2} \approx 4.71 \quad 2\pi \approx 6.28$$

Example 7

In which quadrant does the terminal side of an angle of 3 radians lie?

Homework: Angles and Radian/Degree Measure worksheet.

4.2 Linear and Angular Speed**Objectives**

- Define one radian using the arc-length-to-radius relationship.
- Apply the arc length formula $s = \theta r$ and sector area formula $A = \frac{1}{2}\theta r^2$.
- Distinguish between linear and angular speed in circular motion.
- Solve applied problems involving rotating objects.

Radian Measure

A _____ is the measure of a central angle whose intercepted arc has a length equal to the radius of the circle. We use the letter _____ for arc length.

This definition comes from the geometric proportion:

$$\frac{\text{arc length}}{\text{circumference}} = \frac{\text{central angle}}{2\pi} \implies \frac{s}{2\pi r} = \frac{\theta}{2\pi} \implies \frac{s}{r} = \theta$$

Fact

Arc Length Formula

The arc length s intercepted by a central angle θ in a circle of radius r is:

$$s = \theta r$$

where θ is the angle measured in _____.

Example 1

A circle has radius $r = 4$ inches. Find the length of the arc intercepted by a central angle of 240° .

Fact

Area of a Sector

The area A of a sector of a circle of radius r with a central angle θ is:

$$A = \frac{1}{2} \theta r^2$$

where θ is the angle measured in _____.

Linear and Angular Speed

When a point travels around a circle of radius r at a constant rate:

- _____ (ω) measures how fast the _____ changes:

$$\omega = \frac{\theta}{t} \quad (\text{typically in radians per unit time})$$

- _____ (v) measures how fast the point moves along the _____:

$$v = \frac{s}{t} \quad (\text{typically in length units per unit time})$$

These two speeds are related by the formula:

$$v = r \omega$$

Example 2

The second hand of a clock is 10.2 cm long. Find the linear speed (in cm/sec) of its tip.

Example 3

The blades of a wind turbine are 116 ft long. The propeller rotates at 15 revolutions per minute. Find:

- (a) the angular speed in rad/min.
- (b) the linear speed of the tip of a blade in ft/min.

Example 4

A wheel with diameter 24 inches rotates at 800 rpm. Find:

- (a) the angular speed.
- (b) the linear speed of a point on the circumference in miles per hour (mph). (5280 ft = 1 mile).

Homework: Linear and Angular Speed worksheet.

4.3 Trigonometric Functions: The Unit Circle

Objectives

- Derive coordinates for special angles on the unit circle using special right triangles.
- Write the six trigonometric functions of any standard angle on the unit circle.
- State and apply periodic properties and even/odd symmetry of trigonometric functions.
- Simplify expressions using even/odd identities.
- Evaluate trigonometric functions at non-standard radian values with a calculator.

Special Right Triangles Refresher

The two special right triangles supply the side ratios for every special angle on the unit circle:

- **45-45-90 Triangle:** Legs are n , and the hypotenuse is _____.
- **30-60-90 Triangle:** Short leg is n , long leg is _____, and the hypotenuse is _____.

The Unit Circle

The _____ is a circle of radius $r = 1$ centered at the origin.

For any angle θ in standard position, let $P(x, y)$ be the point where the terminal side intersects the unit circle. Drop a perpendicular to the x -axis to form a right triangle with hypotenuse 1.

- At $\theta = \frac{\pi}{6}$ (30°), the coordinates are _____.
- At $\theta = \frac{\pi}{4}$ (45°), the coordinates are _____.
- At $\theta = \frac{\pi}{3}$ (60°), the coordinates are _____.

Unit Circle Definitions

Since the hypotenuse $r = 1$, the six trigonometric functions of θ are defined as:

$$\begin{aligned} \sin \theta &= y & \cos \theta &= x & \tan \theta &= \frac{y}{x} \quad (x \neq 0) \\ \csc \theta &= \frac{1}{y} \quad (y \neq 0) & \sec \theta &= \frac{1}{x} \quad (x \neq 0) & \cot \theta &= \frac{x}{y} \quad (y \neq 0) \end{aligned}$$

Fact

Signs of Trigonometric Functions

The mnemonic _____ (“All Students Take Calculus”) tells us which trig functions (along with their reciprocals) are positive in each quadrant:

- **Quadrant I (A):** _____ functions are positive.
- **Quadrant II (S):** Only _____ (and cosecant) are positive.
- **Quadrant III (T):** Only _____ (and cotangent) are positive.
- **Quadrant IV (C):** Only _____ (and secant) are positive.

Example 1

Write down the exact values of the six trigonometric functions of $\theta = \frac{5\pi}{4}$.

Example 2

Write down the exact values of the six trigonometric functions of $\theta = -\frac{\pi}{3}$.

Periodicity

Trigonometric functions repeat their values at regular intervals.

- Sine, Cosine, Secant, Cosecant repeat every _____ radians:

$$\sin(\theta + 2\pi k) = \sin \theta \quad \cos(\theta + 2\pi k) = \cos \theta \quad \text{for } k \in \mathbb{Z}$$

- Tangent and Cotangent repeat every _____ radians:

$$\tan(\theta + \pi k) = \tan \theta \quad \cot(\theta + \pi k) = \cot \theta \quad \text{for } k \in \mathbb{Z}$$

Example 3

Evaluate $\sin\left(\frac{13\pi}{6}\right)$ and $\cos\left(\frac{13\pi}{6}\right)$.

Example 4

Evaluate with a calculator. Round to three decimal places.

a) $\sin\left(\frac{2\pi}{3}\right)$

b) $\cot 1.5$

Even and Odd Trigonometric Functions

Functions can be classified by their symmetry properties:

- **Even Functions** satisfy $f(-\theta) = f(\theta)$. These are _____ and _____.

$$\cos(-\theta) = \cos \theta \quad \sec(-\theta) = \sec \theta$$

- **Odd Functions** satisfy $f(-\theta) = -f(\theta)$. These are _____, _____, _____, and _____.

$$\sin(-\theta) = -\sin \theta \quad \tan(-\theta) = -\tan \theta$$

Example 5

Show that $\sin(-x) \tan(-x) + \cos(-x) = \sec(x)$.

Homework: Trigonometric Functions: The Unit Circle worksheet.

4.4 Trigonometric Functions of Any Angle

Objectives

- Find the reference angle θ' for any angle θ .
- Find the six trigonometric functions of an angle given a point on its terminal side.
- Find other trigonometric values given one ratio and a quadrant or sign constraint.
- Find all angles in a specified interval satisfying a given trigonometric equation using reference angles.

Reference Angle θ'

The _____ θ' is the positive acute angle formed by the terminal side of θ and the x -axis.

To find θ' for a positive angle $\theta < 360^\circ$ (or $\theta < 2\pi$), use the quadrant formulas:

- **Quadrant I:** _____
- **Quadrant II:** _____ or _____
- **Quadrant III:** _____ or _____
- **Quadrant IV:** _____ or _____

Example 1–3

Find the reference angle θ' for each given angle.

1) $\theta = 330^\circ$

2) $\theta = -\frac{5\pi}{6}$

3) $\theta = 2.3$ (radians)

Trig Functions of Any Angle

Let θ be an angle in standard position. Let $P(x, y)$ be any point on the terminal side of θ (other than the origin), and let r be the distance from the origin to P :

$$r = \sqrt{x^2 + y^2}$$

The six trigonometric functions of θ are defined as:

$$\begin{aligned} \sin \theta &= \frac{y}{r} & \cos \theta &= \frac{x}{r} & \tan \theta &= \frac{y}{x} \quad (x \neq 0) \\ \csc \theta &= \frac{r}{y} \quad (y \neq 0) & \sec \theta &= \frac{r}{x} \quad (x \neq 0) & \cot \theta &= \frac{x}{y} \quad (y \neq 0) \end{aligned}$$

Note that r is a distance, so it is _____ ($r > 0$).

Example 4

The terminal side of an angle θ in standard position contains the point $(8, -15)$. Find the exact values of all six trigonometric functions of θ .

Example 5

Given $\sin \theta = -\frac{2}{3}$ and $\tan \theta > 0$, find $\cos \theta$ and $\cot \theta$.

Note**Solving for Angles**

To solve a trigonometric equation like $\sin \theta = c$ or $\cos \theta = c$ for all angles in one revolution:

1. Find the reference angle θ' by taking the inverse trig function of the _____ of c .
2. Use _____ to find the two quadrants where the function has the correct sign.
3. Build the two angles from θ' using the formulas:

$$\text{QII: } 180^\circ - \theta' \quad \text{QIII: } 180^\circ + \theta' \quad \text{QIV: } 360^\circ - \theta'$$

Example 6–7

Solve each trigonometric equation on the given interval.

6) If $\tan \theta = -0.4623$ and $0^\circ \leq \theta < 360^\circ$, find θ to the nearest 0.1° .

7) If $\cos \theta = -0.3842$ and $0 \leq \theta < 2\pi$, find θ to the nearest 0.0001 radian.

Homework: Trigonometric Functions of Any Angle worksheet.

4.5 Right Triangle Trigonometry

Objectives

- Find the values of the six trigonometric functions of an acute angle in a right triangle given side lengths.
- Apply reciprocal, quotient, Pythagorean, and cofunction identities.
- Convert between degrees and degrees-minutes-seconds (DMS) notation.
- Solve real-world right triangle applications (angles of elevation and depression).

Right Triangle Definitions (SOH-CAH-TOA)

For an acute angle θ in a right triangle, we define the six trigonometric ratios as:

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} \quad \cos \theta = \frac{\text{adj}}{\text{hyp}} \quad \tan \theta = \frac{\text{opp}}{\text{adj}}$$

$$\csc \theta = \frac{\text{hyp}}{\text{opp}} \quad \sec \theta = \frac{\text{hyp}}{\text{adj}} \quad \cot \theta = \frac{\text{adj}}{\text{opp}}$$

where “opp” is the side opposite θ , “adj” is the side adjacent to θ , and “hyp” is the hypotenuse.

Example 1

In right triangle $\triangle ABG$, the right angle is at B , with $AB = 21$, $BG = 28$, and $AG = 35$. Find the six trigonometric functions of angle G .

Example 2

Given $\cos \theta = \frac{7}{8}$, find the exact values of the other five trigonometric ratios of θ (assume θ is acute).

Example 3

Use an equilateral triangle of side length 2 to find the exact values of $\sin 60^\circ$, $\cos 60^\circ$, $\sin 30^\circ$, and $\cos 30^\circ$.

Cofunction Identities

Cofunctions of complementary angles are equal.

$$\sin \theta = \cos(90^\circ - \theta) \quad \tan \theta = \cot(90^\circ - \theta) \quad \sec \theta = \csc(90^\circ - \theta)$$

$$\cos \theta = \sin(90^\circ - \theta) \quad \cot \theta = \tan(90^\circ - \theta) \quad \csc \theta = \sec(90^\circ - \theta)$$

Example 4

Let θ be an acute angle with $\tan \theta = \frac{1}{3}$. Find $\cot \theta$ and $\sec \theta$. Show both methods: reference triangle vs. identities.

Degrees Minutes and Seconds (DMS)

Like time, degrees can be divided into _____ (') and _____ (") :

$$1^\circ = 60' \quad 1' = 60'' \quad \Rightarrow \quad 1^\circ = 3600''$$

To convert $D^\circ M'S''$ to decimal degrees:

$$\text{Decimal Degrees} = D + \frac{M}{60} + \frac{S}{3600}$$

Example 5

Evaluate $\sec(9^\circ 40' 12'')$ using a calculator. Round to three decimal places.

Angle of Elevation and Depression

- An _____ is measured upward from the horizontal.
- An _____ is measured downward from the horizontal.

Example 6

A mom 3.5 ft off the floor sees a pen at an angle of depression of 65° . How far away is the pen along the floor from directly below her eyes? Round to the nearest hundredth of a foot.

Example 7

From a point 200 ft from the base of a building, the angle of elevation to the bottom of a smokestack on top of the building is 35° and to the top of the smokestack is 53° . Find the height of the smokestack alone. Round to the nearest tenth of a foot.

Homework: Right Triangle Trigonometry worksheet.